Do Simple Technical Trading Rules Survive Realistic Costs?

Walk-Forward, Deflated-Sharpe and Monte-Carlo Evidence from 30 Years of Equity-Index Data

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Abstract

We ask whether common rule-based technical strategies deliver *economically and statistically* significant risk-adjusted returns, net of realistic trading and financing costs, on liquid equity indices. Using walk-forward analysis, a contract-for-difference (CFD) financing model, an explicit leverage analysis, and a battery of significance tests — annualised and *deflated* Sharpe ratios, bootstrap confidence intervals and Monte-Carlo trade resampling — we evaluate moving-average, breakout-trend, RSI(2) mean-reversion and opening-range-breakout rules on the S&P 500 and ASX 200 (daily, 1996–2026) and on 5-minute S&P 500 data (2024–2026). On total return, every rule underperforms buy-and-hold. On a risk-adjusted basis the picture is subtler and more interesting: a daily breakout-trend rule on the S&P 500 attains a higher Sharpe ratio (0.60 vs 0.49) and a far smaller maximum drawdown (–4.9% vs –57%) than buy-and-hold, and this effect survives correction for multiple testing (deflated Sharpe ratio 0.98; 99% of bootstrap resamples profitable). However, the same rule's absolute return is economically negligible (CAGR below 1%, about 0.4% after financing), it does *not* replicate significantly on the ASX 200 (deflated Sharpe ratio 0.45), and leverage cannot monetise the low drawdown because CFD financing scales with it. Mean-reversion and intraday breakout rules show no significant edge (deflated Sharpe ratio ≈ 0). The practical implication for retail traders is direct: simple technical rules do not produce exploitable excess returns net of costs on these markets; the only robust effect is risk reduction with near-zero return, consistent with market efficiency [4, 5]. Our contribution is as much methodological as empirical — a reproducible, overfitting-resistant validation pipeline that distinguishes genuine edge from backtest artefact [1, 2, 3].

1. Research Contributions

This paper makes four contributions:

- **1. Cost- and financing-realistic evaluation.** We model per-trade spread and slippage *and* CFD overnight financing on the full notional — the cost that most retail backtests omit and that proves decisive for any rule held overnight.
- **2. Overfitting-resistant validation.** Every result is out-of-sample under walk-forward analysis, judged by a deflated Sharpe ratio that corrects for the number of parameter trials, bootstrap confidence intervals, and Monte-Carlo trade resampling — not a single in-sample backtest.
- **3. An explicit leverage analysis.** We show analytically and empirically why a low-drawdown strategy's apparent leverage-efficiency fails for overnight CFD positions: financing scales with leverage, so the risk-adjusted ratio degrades.
- **4. A fully reproducible research lab.** All experiments run from single-command presets over a cached data store, with a vectorised O(n) backtest engine; the methodology transfers directly to other markets and rules.

2. Data

Daily OHLC series for the S&P 500 and ASX 200 span 24 June 1996 to June 2026 (7,545 and 7,581 bars), covering the dot-com crash, the 2008 global financial crisis, the 2020 COVID shock and the 2022 bear market. Intraday data are 5-minute bars for the SPY ETF, 94,384 bars over June 2024–June 2026 (538 sessions). All series are price indices (dividends excluded), which understates the buy-and-hold benchmark and therefore biases our conclusions conservatively against the strategies. Index CFDs on the same underlyings are the traded instruments.

3. Methodology

Walk-forward analysis. A window is rolled forward: on each in-sample (train) window we grid-search parameters, select the best, freeze them, and trade them on the next unseen out-of-sample (test) window, compounding across folds into a continuous out-of-sample track the optimiser never saw. Daily runs use a 2,500-bar (~10y) train and 750-bar (~3y) test; intraday runs use 15,000/5,000.

Costs, financing, leverage. Each trade pays spread plus slippage. Overnight CFD positions pay financing on the full notional (modelled at ~5%/yr, a mid-cycle estimate), prorated by holding period; intraday rules flat by session close pay none. Leverage is modelled as position-size scaling. A time-stop caps holding length; an intraday session-flat rule prevents overnight exposure.

Statistical tests. We report the annualised Sharpe ratio of the out-of-sample equity curve and the buy-and-hold benchmark; the *deflated Sharpe ratio* (DSR) [1], which gives the probability the Sharpe ratio is genuinely positive after correcting for the number of parameter configurations tried; a t-test on mean per-trade return; bootstrap 5th–95th-percentile confidence intervals on total return; and Monte-Carlo resampling of the trade sequence to obtain the distribution of terminal return and maximum drawdown. A DSR above 0.95 is treated as significant.

Strategies. (i) moving-average crossover; (ii) breakout-trend (a close-based Donchian breakout filtered by a long trend moving average); (iii) RSI(2) mean-reversion [10] with both ATR-target [12] and reversion-to-MA exits; (iv) opening-range breakout (trade the break of the first 30–60 minutes of each session, flat by close).

4. Results

4.1 Performance and risk

Configuration	OOS ret %	CAGR %/yr	Sharpe	Max DD %	MAR	Tr.	B&H ret %
Breakout-trend — S&P 500, daily	+18.8	0.97	0.60	-4.9	0.20	99	+298
Breakout-trend — ASX 200, daily	+4.7	0.26	0.17	-5.0	0.05	100	+46
Breakout-trend — ASX 200, hourly	-5.5	—	—	-6.3	—	34	—
Mean-reversion (RSI-2) — S&P 500, daily	-6.0	—	—	-7.6	—	148	+298
Mean-reversion (RSI-2) — SPY, 5-min	-8.7	—	—	-9.9	—	838	+27
Opening-range breakout — SPY, 5-min	-19.1	—	—	-23.7	—	5632	+27

Table 1. Out-of-sample performance. Sharpe is annualised; MAR = CAGR / |max drawdown|; Tr. = trades; B&H = buy-and-hold total return. Buy-and-hold Sharpe / MAR were 0.49 / 0.14 (S&P 500) and 0.21 / 0.04 (ASX 200). Returns exclude financing; see Table 3.

Only the S&P 500 daily breakout-trend rule attains a Sharpe ratio (0.60) and MAR (0.20) above buy-and-hold (0.49 / 0.14), achieved through a maximum drawdown near 5% versus 57% for the index. Yet it captures under one-tenth of the index's total appreciation (Figure 1) and earns under 1% per year.

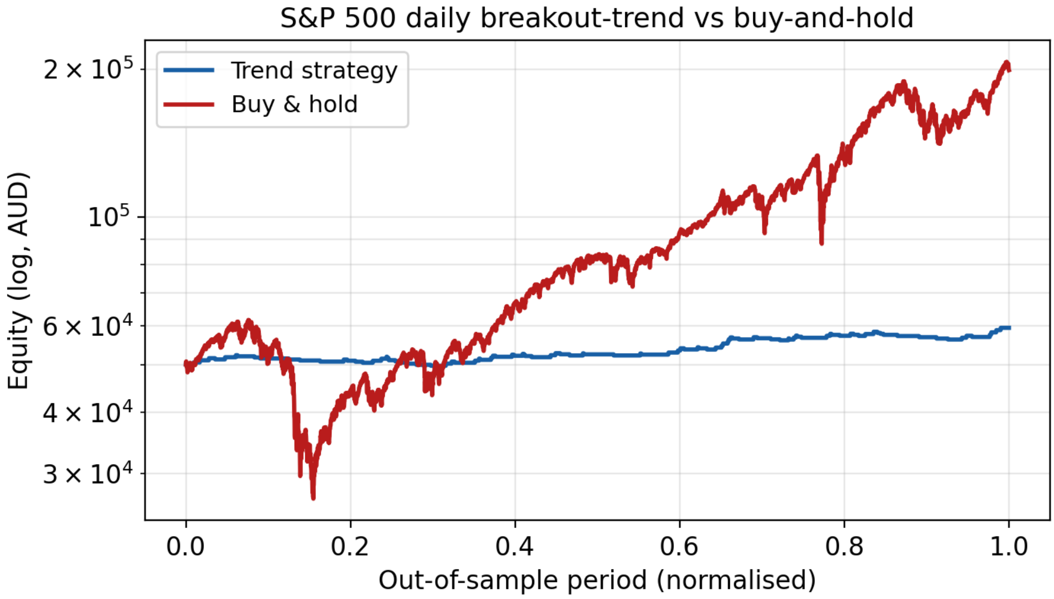

Figure 1. S&P 500 daily breakout-trend versus buy-and-hold over the out-of-sample period (log scale). The strategy sidesteps the 2008 drawdown but captures little long-run appreciation.

4.2 Statistical significance

Configuration	Sharpe (strat/B&H)	t	p	DSR	Bootstrap 90% CI, ret %	Sig.?
Breakout-trend — S&P 500, daily	0.60 / 0.49	2.35	0.019	0.98	[+5.3, +33.9]	Yes
Breakout-trend — ASX 200, daily	0.17 / 0.21	0.66	0.51	0.45	[-6.8, +18.0]	No
Mean-reversion — SPY, 5-min	<0	—	—	0.00	[-12.9, -4.4]	No

Table 2. Significance tests. DSR = deflated Sharpe ratio (probability the Sharpe is truly positive after correcting for the nine parameter trials); DSR > 0.95 = significant. Bootstrap CI is the 5th–95th percentile of resampled total return.

The S&P 500 trend result is statistically significant: a deflated Sharpe ratio of 0.98 means the edge is unlikely to be a multiple-testing artefact, and 99% of bootstrap resamples are profitable. Critically, however, the *same* rule on the ASX 200 is indistinguishable from noise (DSR 0.45, p = 0.51), and the intraday mean-reversion rule has a negative Sharpe and DSR of zero. Statistical significance on one index that fails to replicate on a second is weak evidence of a durable, transportable edge.

4.3 Risk versus return, and leverage

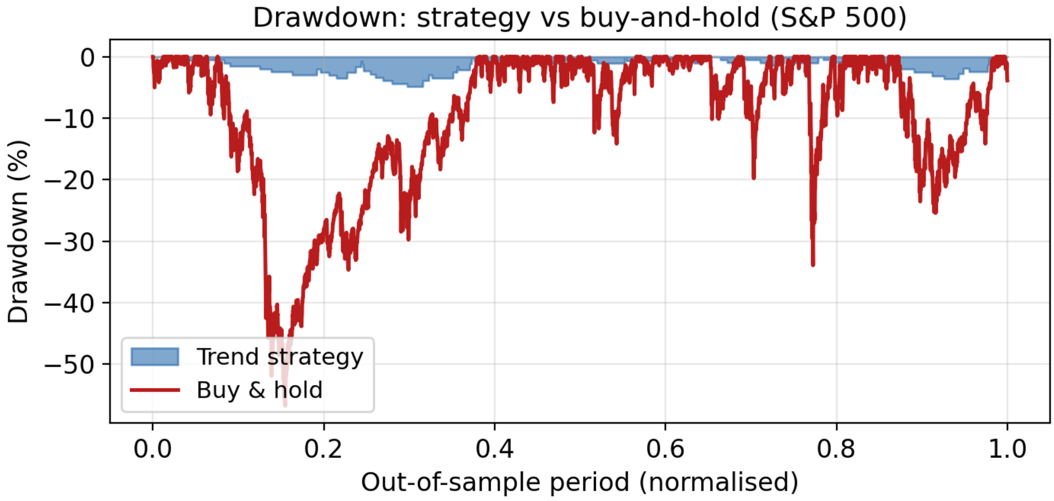

Figure 2. Drawdown paths. The trend rule's worst drawdown is ~5%; buy-and-hold endures ~57% in 2008. This is the strategy's only genuine advantage.

The trend rule sacrifices roughly 279 percentage points of total return (298% → 19%) to cut maximum drawdown from 57% to 5%. A natural question is whether leverage can convert that low drawdown into a competitive return. It cannot: because CFD financing accrues on the full notional, it scales with leverage, so gross return and financing rise together while drawdown grows faster — the return-to-drawdown ratio *falls* with leverage (Table 3).

Leverage	CAGR %/yr	Max DD %	MAR
1×	0.4	-4.5	0.09
5×	1.5	-21.4	0.07
10×	1.9	-38.9	0.05
Index fund (buy & hold)	8.0	-57	0.14

Table 3. Leverage sweep, S&P 500 daily breakout-trend, ~5%/yr financing applied (hence CAGR below the financing-free Table 1 figure). Leverage worsens the risk-adjusted result.

4.4 Parameter stability

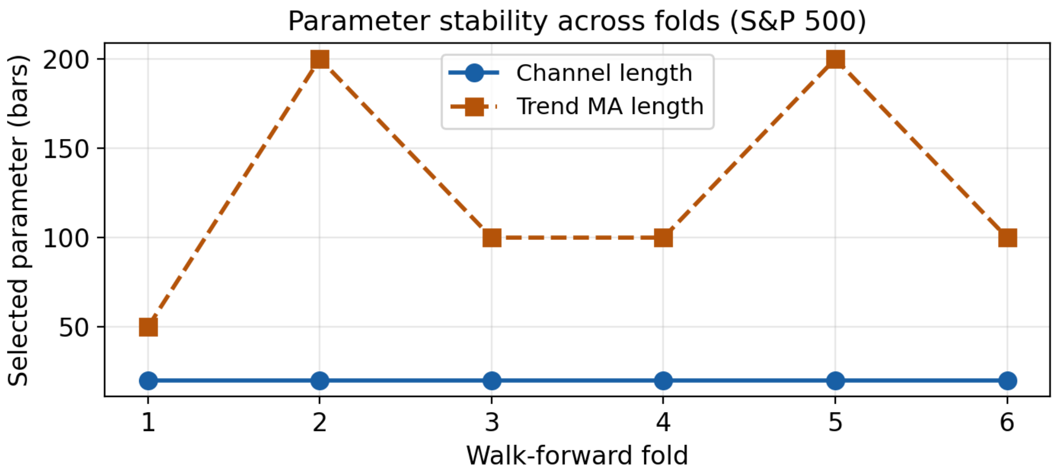

Figure 3. Parameters selected per walk-forward fold (S&P 500). The channel length is identical (20) in all six folds; the trend filter varies modestly (50–200). Stable selections are necessary, though not sufficient, for a non-spurious signal — contrast the hourly and intraday rules, whose selections jumped between extremes.

4.5 Monte-Carlo robustness

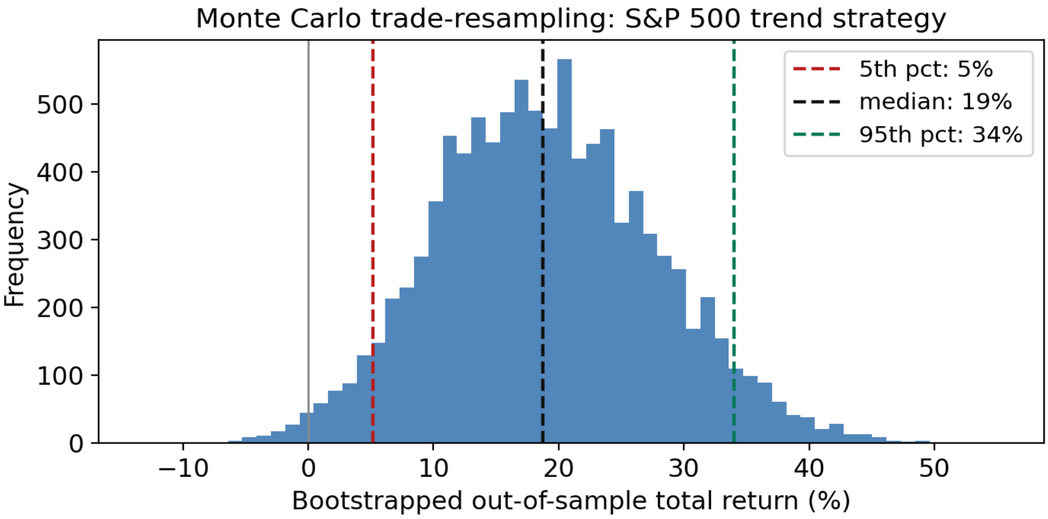

Figure 4. Distribution of out-of-sample total return under 10,000 trade-sequence resamplings (S&P 500 trend). Median +18.7%, 90% interval [+5.3%, +33.9%], 99% of paths profitable; the resampled maximum-drawdown distribution has a median of –3.7% and a 5th-percentile (worst) of –6.9%. The positive return is robust — but small, and well below buy-and-hold in every resample.

5. Discussion

The results are coherent and, with hindsight, expected. Equity indices are among the most liquid and most heavily arbitraged instruments in existence; a durable edge from public, price-only rules would be competed away [4, 5]. Early evidence of profitable technical rules [6] did not survive corrections for data-snooping [3, 2], and our findings echo that arc with modern, cost-aware tooling.

Why trend-following 'works' yet does not pay. Trend rules behave exactly as theory predicts: they exit during sustained declines and so avoid deep drawdowns, at the cost of sitting in cash through much of the recovery. On a single index this trade-off is unattractive — surrendering the overwhelming majority of the return to avoid a drawdown is rational only for an investor who cannot tolerate the drawdown at all. The documented economic value of trend-following comes from applying it across *dozens* of weakly-correlated markets, where the diversified stream of small edges compounds [7, 8]; related momentum effects are well documented cross-sectionally [9]. One index cannot supply that diversification.

Why mean-reversion and breakout fail under costs. RSI(2) mean-reversion wins frequently (high hit-rate) but its average win is smaller than its average loss once spread, slippage and (for overnight variants) financing are charged; its in-sample objective was already negative in most folds. Opening-range breakout trades thousands of times, so transaction costs alone are decisive. Neither edge, if it ever existed, survives realistic frictions.

The leverage trap. The intuition that a low-drawdown strategy is 'leverage-efficient' holds only when the edge per unit of financing is high. Here it is not: financing consumes the thin edge and grows with the very leverage intended to exploit it. The one structurally favourable setting — intraday rules that avoid overnight financing entirely — produced no profitable rule in our tests, though the intraday sample is short.

6. Limitations

'No exploitable edge found' is not 'no edge exists'. (i) Only two indices and a limited family of rules and parameters were tested; (ii) financing is a flat ~5%/yr, whereas the true rate ranged from near zero (2009–2021) to ~7%, which shifts levels though not the ranking; (iii) intraday data span only two years; (iv) price indices exclude dividends, understating buy-and-hold; (v) fills are modelled at next-bar prices with fixed costs, so real slippage, gaps and partial fills are only approximated; (vi) the deflated Sharpe ratio corrects for the trials we ran, not for the implicit trials embedded in choosing the rule families themselves [1, 11].

7. Future Research

- **Diversified futures.** Apply the trend rules across dozens of liquid futures (equity indices, rates, commodities, FX) — the setting where trend-following is documented to pay [7, 8] — and test whether the diversified Sharpe clears costs.
- **Multi-asset portfolios and volatility targeting.** Combine weakly-correlated return streams and size positions to a volatility target, which may improve risk-adjusted return where single-asset rules cannot.
- **Regime-switching overlays.** Use the trend signal not to trade in and out but as a *defensive overlay* on a buy-and-hold core — hold the index, de-risk only in confirmed downtrends — to retain most of the return while cutting tail drawdowns.
- **Machine learning, with discipline.** Feature-based models may capture non-linear structure, but financial ML overfits readily; any such work must use purged/embargoed cross-validation and deflated performance metrics [11].
- **More markets and longer intraday history.** Extend the daily tests to the Nasdaq 100, DAX, FTSE 100 and Nikkei 225, and obtain multi-year 5-minute data, to test breadth and replication directly.

8. Conclusion

Across trend, mean-reversion and breakout rules, on daily, hourly and 5-minute data, evaluated out-of-sample with realistic costs, financing and leverage and a full battery of significance tests, no simple technical strategy delivered economically exploitable excess returns on the S&P 500 or ASX 200. The single statistically significant result — a daily trend rule with a higher Sharpe ratio than buy-and-hold — is a drawdown-reduction effect with near-zero absolute return that leverage cannot monetise and that does not replicate across markets. The evidence is consistent with market efficiency on highly arbitraged instruments. The enduring contribution is the validation methodology: a reproducible, overfitting-resistant pipeline that reports the truth cheaply, so capital is never committed to a strategy whose apparent edge is an artefact. The most important finding is not that these particular strategies failed, but that a rigorous validation process identified their limitations before capital was committed. In practical terms, the validation framework may be more valuable than any individual trading rule examined.

9. Publication Roadmap

This manuscript is positioned as a working paper. A practical route to dissemination is: (i) post the current version as a preprint on **SSRN** and **arXiv** (q-fin.ST / q-fin.TR) for citability and feedback; (ii) extend the study with the additional indices and a diversified-futures test (Section 7) and submit the expanded version to a practitioner-facing peer-reviewed outlet such as the *Journal of Investing* or the *Journal of Financial Data Science*. The reproducible code and data pipeline accompany the paper as supplementary material.

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Appendix: Reproducibility

All experiments run from single-command presets over a cached data store, e.g. `python -m scripts.run_walkforward spx`. The pipeline comprises a historical-data cache; an event-driven backtester with vectorised, causal signal computation (O(n) replay) and cost/financing/leverage/time-stop and intraday session-flat logic; a walk-forward engine with parameter-stability reporting; and the statistical battery of Section 3.